



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2016
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
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H2 BIOLOGY

9648/02

Paper 2 Core Paper

16 September 2016

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Answer **one** question on the answer paper provided.

At the end of the examination,

1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
 - Section A (Part I)
 - Section A (Part II)
 - Section B

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 10
2	/ 9
3	/ 10
4	/ 11
5	/ 11
6	/ 9
7	/ 11
8	/ 9
Section B	
9 / 10	/ 20
Total	/ 100

This paper consists of **20** printed pages.

[Turn over]

Section A (Part I)
Answer **all** the questions in this section.

*For
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QUESTION 1

The structure of the tubulin dimer, the protein that forms microtubules by polymerisation, is shown in **Fig. 1.1**.

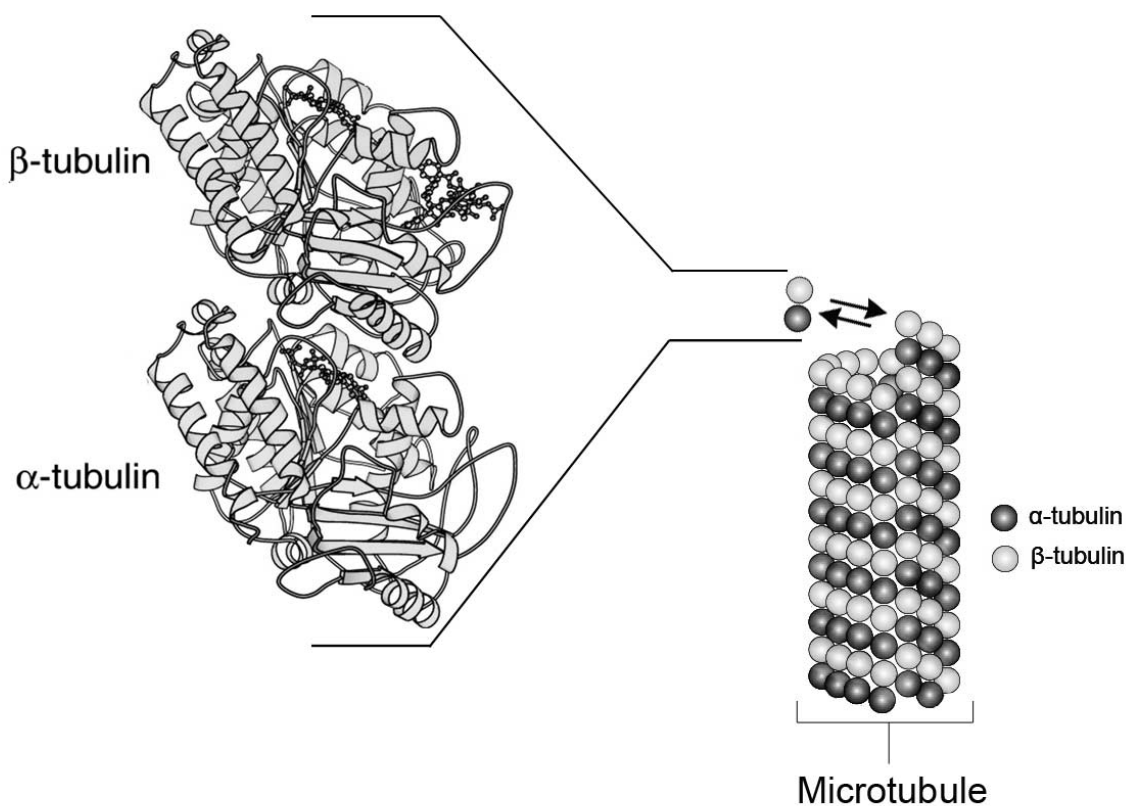


Fig. 1.1

- (a)** With reference to **Fig. 1.1**, name the secondary structures present in tubulin. [1]

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Tubulin inhibitors like paclitaxel and vinblastine have been utilised in chemotherapy drug trials to treat cancers. All tubulin inhibitors are known to bind to the β -tubulin subunit.

- (b)** Explain how tubulin inhibitors reduce tumour formation. [3]

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A trial was conducted to compare the effects of vinblastine and paclitaxel on the SKOV3 ovarian cancer cell line and the PC3 prostate cancer cell line.

Table 1.1 below shows the results of the trial. The researcher measured the number of months in which the mass of tumours increased to critical mass after treatment with vinblastine and paclitaxel.

	No. of months in which the mass of tumours increased to critical mass	
	SKOV3	PC3
Untreated	0.5	1.0
Vinblastine	5.7	10.0
Paclitaxel	10.1	9.1

Table 1.1

- (c) With reference to **Table 1.1**, compare the effects of vinblastine and paclitaxel on tumour growth in the two cancer cell lines. [4]

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- (d) Suggest and explain why different tumour cells may exhibit different levels of resistance to the same drug. [2]

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[Total: 10]

QUESTION 2

Fig. 2.1 shows the process of translation.

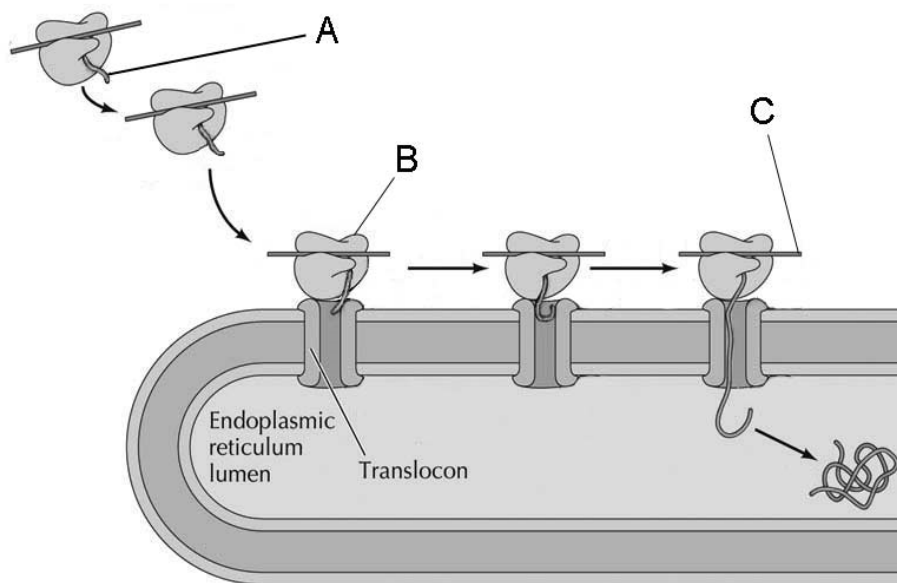


Fig. 2.1

(a) (i) Label structures **A**, **B** and **C**.

[3]

A

B

C

(ii) Suggest the role of the translocon in protein synthesis.

[1]

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(b) List two ways in which transcription differs from DNA replication.

[2]

1.

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2.

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(c) Explain how complementary base pairing facilitates the storage and transmission of genetic information. **[3]**

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[Total: 9]

QUESTION 3

The measles virus (MV) is a spherical, non-segmented, single-stranded negative sense RNA virus. The structure of MV is shown in **Fig. 3.1**.

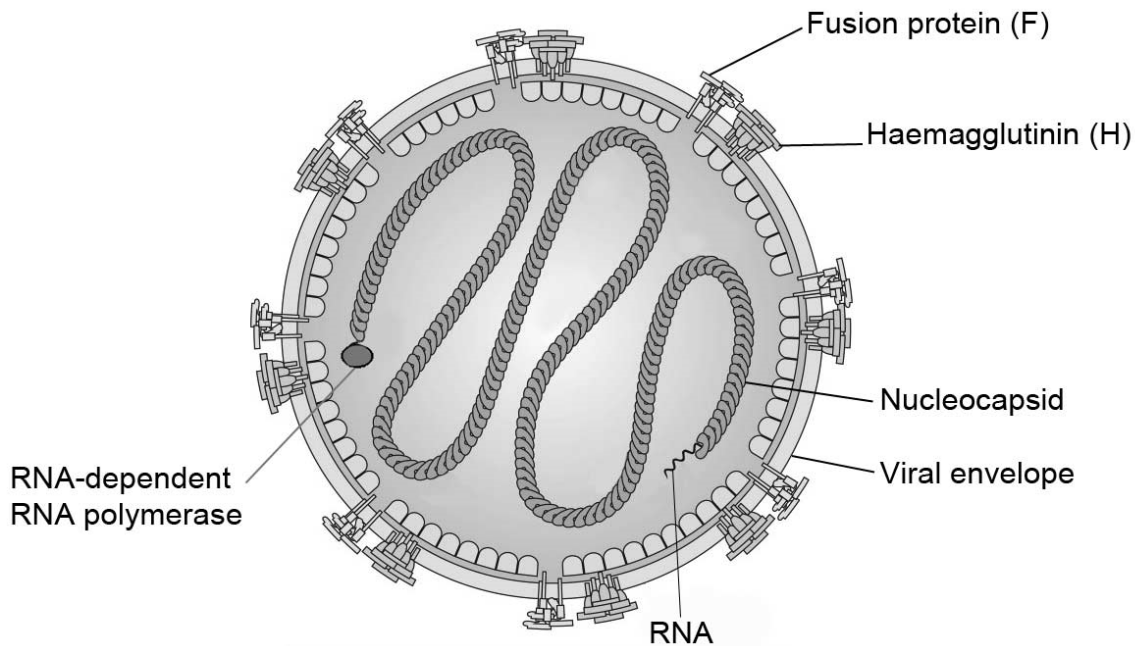


Fig. 3.1

(a) With reference to **Fig. 3.1**, describe two structural differences between MV and HIV. [2]

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2.
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MV only infects cells that have a membrane glycoprotein known as signaling lymphocyte activation molecule (SLAM). When MV infects a cell, **H** acts before **F**. After the virus binds to the host cell, only the nucleoprotein with the viral polymerase enters the host cell and the virus is replicated.

(b) With reference to **Fig. 3.1** and the information provided, suggest how MV infects a cell with SLAM glycoproteins. [3]

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Both MV and HIV infect cells of the immune system. Upon infection, MV causes highly contagious measles which is an airborne disease spreads through the coughs and sneezes of those infected.

(c) (i) Explain how HIV infection causes diseases. [4]

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(ii) Suggest why MV is transmitted at a faster rate as compared to HIV. [1]

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[Total: 10]

QUESTION 4

- (a) Telomerase is a ribonucleoprotein which comprises telomerase reverse transcriptase (TERT) protein and telomerase RNA.

Outline how telomerase is formed.

[4]

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During human embryonic development, telomerase activity is activated in embryonic stem cells to enable high proliferation rate of the cell. However, the telomerase activity is usually diminished after birth and the level of telomerase activity is absent in most of the somatic cells.

Fig. 4.1 shows the *TERT* promoter in the two types of cells.

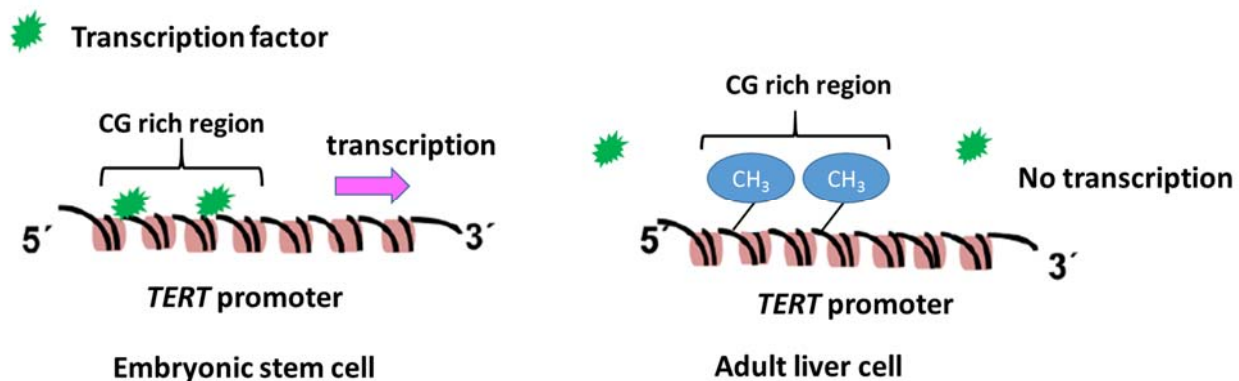


Fig. 4.1

- (b) With reference to Fig. 4.1 and your knowledge, explain why telomerase activity is absent in adult liver cells.

[3]

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(c) The process occurring in adult liver cells shown in **Fig. 4.1** also occurs in prokaryotic cells.

State how the outcome of the process in prokaryotes differs from that in adult liver cells. [1]

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(d) Outline the roles of telomeres in eukaryotic cells. [3]

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[Total: 11]

QUESTION 5

To study the inheritance of coat colour and eye colour in deer-mice, scientists performed two crosses and the table below shows the phenotypes of the F₁ generations from these two crosses.

Cross	Parents (pure bred)	F ₁ phenotype	Number of F ₁ progeny
1	Black eye, coloured female X Pink eye, albino male	All black eye, coloured mice	77
2	Black eye, coloured male X Pink eye, albino female	All black eye, coloured mice	68

The F₁ generation were then interbred and the following F₂ offspring were produced:

Black eye, coloured	295
Black eye, albino	42
Pink eye, coloured	46
Pink eye, albino	33

(a) Explain the purpose of carrying out crosses 1 and 2.

[2]

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(b) Using suitable symbols, draw a genetic diagram to explain the results of F₁ cross.

[5]

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(c) State the expected ratio of the F₂ phenotypes if Mendelian law applies to the two gene loci. [1]

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The chi-squared (χ^2) test was performed on these results, giving a calculated value for χ^2 of 47.527.

The χ^2 distribution table and equation to calculate χ^2 is shown below.

number of degrees of freedom (v)	probability
	0.05
1	3.84
2	5.99
3	7.82
4	9.49

(d) Using the calculated value of χ^2 and the table of probabilities provided in the table above, explain the conclusions drawn from the (χ^2) test. [3]

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[Total: 11]

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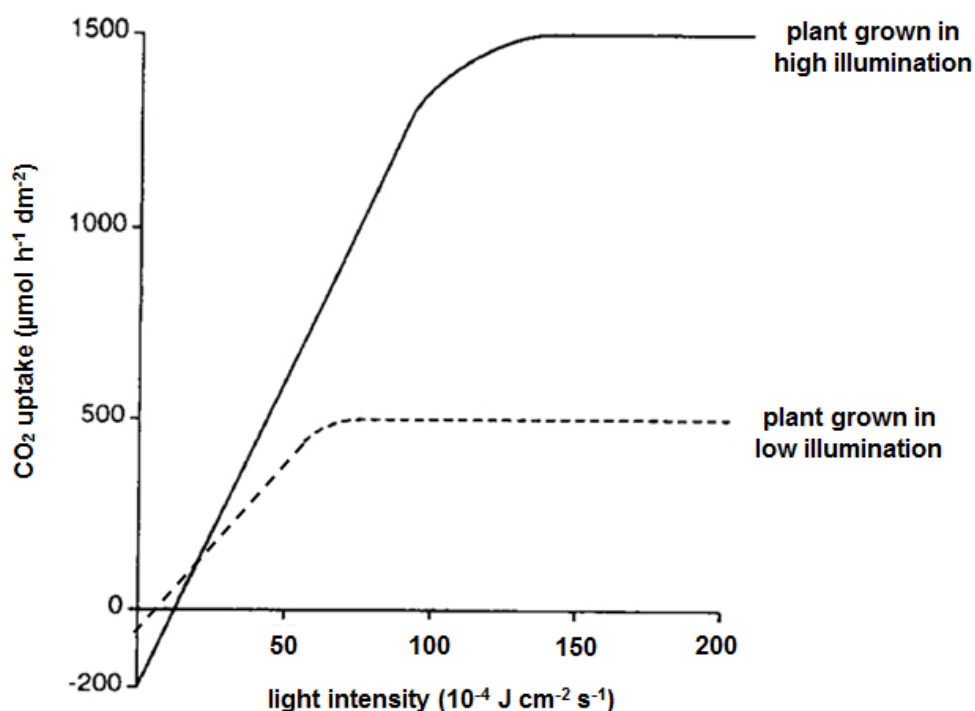
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H2 Biology (9648/02)

Question 6	Question 7	Question 8
/ 9	/ 11	/ 9

Section A (Part II)Answer **all** the questions in this section.**QUESTION 6**

Two groups of white mustard plants, *Sinapis alba*, were grown, one group under high illumination, the other under low illumination. When fully grown, the effect of increasing light intensity on the rate of photosynthesis in the two groups of plants was measured.

Fig. 6.1 shows the results.

**Fig. 6.1**

(a) With reference to **Fig. 6.1**,

- (i)** state and explain the effect of light intensities above $150 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$ on the rate of photosynthesis in plants grown in high illumination. [2]

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(ii) state with evidence, **two** ways in which the carbon dioxide uptake of both plants differ at light intensities below $50 \times 10^{-4} \text{ Jcm}^{-2}\text{s}^{-1}$. [4]

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(b) Outline the fate of each product of photolysis in the light dependent reaction. [3]

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[Total: 9]

QUESTION 7

Fig. 7.1 is an electron micrograph of a section through a myelinated neurone showing the Schwann cell and axon membrane.

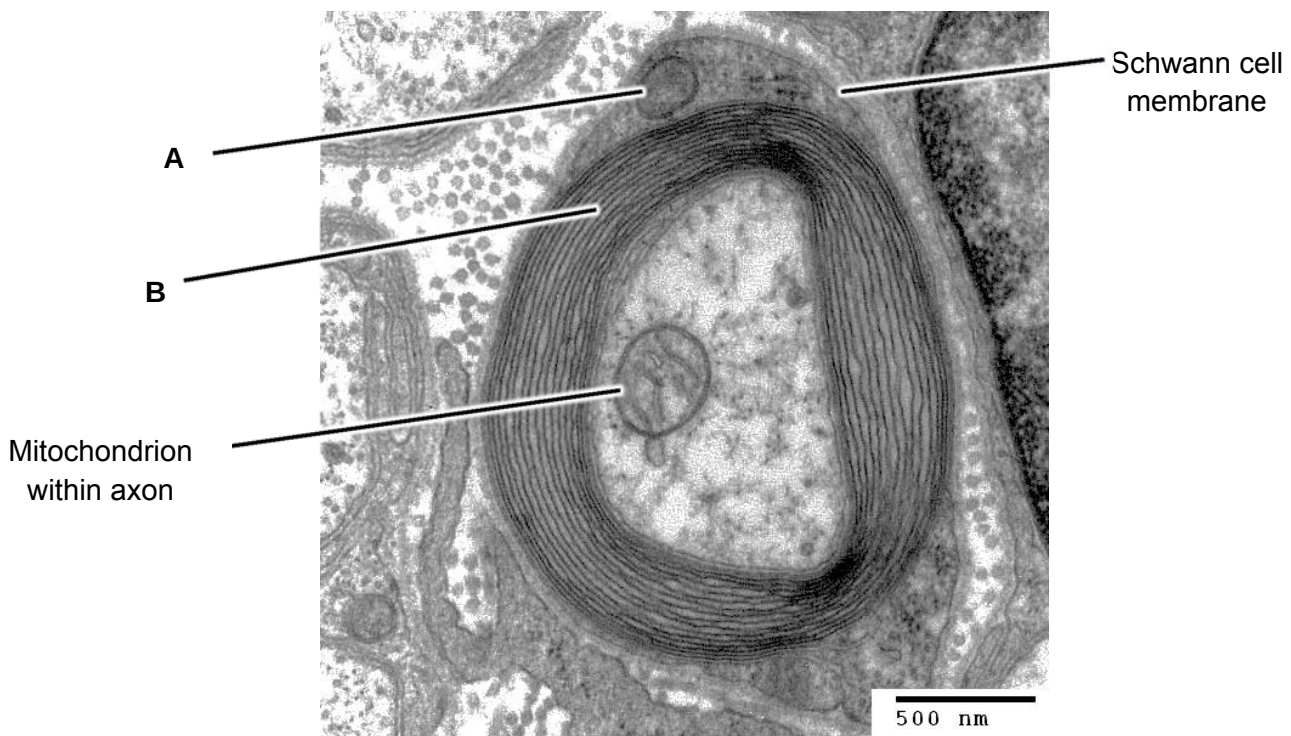


Fig. 7.1

(a) Identify the structures labelled **A** and **B**.

[2]

A

B

(b) **Fig. 7.2** shows the percentage of energy used for various processes involved in the maintenance of resting potentials and in the reception and transmission of action potentials by a neuron.

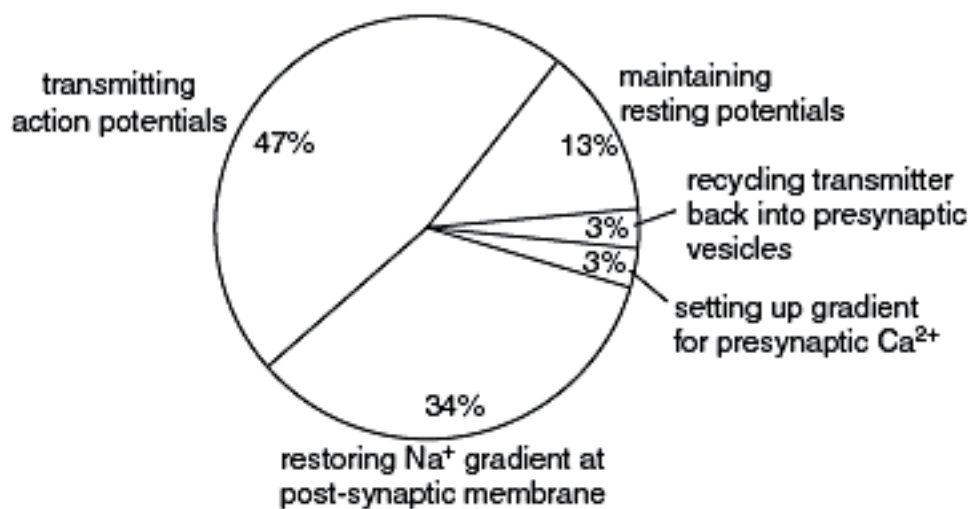


Fig. 7.2

(i) Explain why maintaining a resting potential requires energy. [2]

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(ii) Neurons contain large numbers of mitochondria. There are more mitochondria in each dendrite than in the axon.

With reference to **Fig. 7.2**, suggest reasons for the distribution of mitochondria. [3]

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(c) Describe the role of Ca^{2+} in the passage of impulses across a synapse. [2]

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(d) Synapses slow down the rate of transmission of nerve impulses but serve other important roles in the nervous system.

Outline two roles of synapses in the nervous system. [2]

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2.
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[Total: 11]

QUESTION 8

Mole rats, *Spalax ehrenbergi*, are mammals that live in groups in underground burrows. They are blind, and communicate with each other through sound and scent. Males make a purring call when they attempt to persuade females to mate with them.

In Israel, the mole rats found in different parts of the country all look identical. However, there are actually four different populations with different chromosome numbers, which live in different climatic regions.

Table 8.1 shows the four populations of mole rats and information about the purring calls used by the males in each population. The call of the males were analysed by measuring the number of sound pulses per second, and also the frequencies of the sounds that they made.

Chromosome number of population		52	54	58	60
Climatic region in which population lives		Cool and humid	Cool and dry	Warm and humid	Warm and dry
Purring call made by males	Mean number of pulses per second	21.0	25.3	23.9	23.2
	Mean major frequency/ kHz	595	555	583	562

Table 8.1

Researchers investigated how female mole rats from each of the four populations responded to purring calls made by males from the same population, and by males from different populations.

A female was placed midway between two loudspeakers, and recorded calls from two males were played to her simultaneously. The researchers noted which loudspeaker the female moved towards. This was repeated with many different females from each population.

The results are shown in **Table 8.2**.

Population chromosome number	Percentage of females preferring the purring call of males from their own population
52	79
54	77
58	78
60	44

Table 8.2

- (a) With reference to **Table 8.2**, describe the extent to which female mole rats show a preference for the purring calls of males from their own population. [2]

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- (b) With reference to the data in both **Table 8.1** and **Table 8.2**, discuss whether these four populations of mole rats should be classified as different species. [4]

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The phylogenetic relationship of seven genera of mole rats was investigated using nucleotide sequences of the *12S rRNA* gene obtained from mitochondrial DNA.

Fig. 8.1 shows a phylogenetic tree of the mole rats based on this *rRNA* gene nucleotide sequence data.

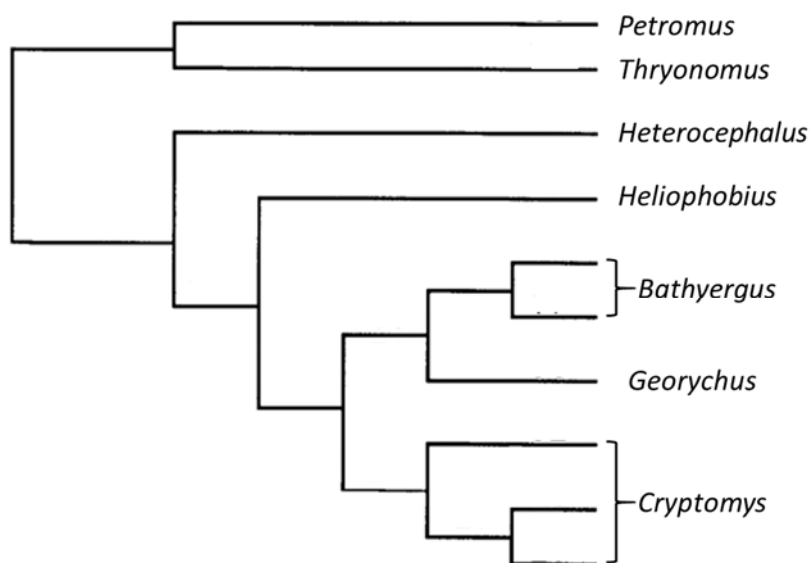


Fig. 8.1

(c) Describe the advantages of using nucleotide data such as the *12S rRNA* gene in classifying the mole rats. [3]

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[Total: 9]

Section B
Answer **one** question.

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Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in questions **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 9

- (a)** Describe the role of vesicles in a cell. [6]
- (b)** Describe one causative factor of cancer and explain how this factor increases the chances of cancerous growth. [6]
- (c)** Describe the differences between the control of gene expression in prokaryotic and eukaryotic cells. [8]

[Total: 20]

QUESTION 10

- (a)** Describe the structure of collagen and how it is related to its function. [8]
- (b)** Explain why antibiotic resistance spreads so rapidly among bacteria. [6]
- (c)** Describe the main differences between glucagon and insulin signalling in liver cells. [6]

[Total: 20]

• END OF PAPER 2 •